

```
(%i1) A:matrix([2,1],[1,2]);
```

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

```
(%i13) eval:eigenvalues(A);
```

```
eval [[3,1],[1,1]]
```

```
(%i14) evvec:eigenvectors(A);
```

```
evvec [[ [3,1],[1,1]], [[ [1,1],[1,-1]] ]]
```

```
(%i15) evvecmatrix:matrix([1,1],[1,-1]);
```

$$\text{evvecmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

```
(%i16) r:rank(evvecmatrix);
```

```
r 2
```

```
(%i17) len:length(evvecmatrix);
```

```
len 2
```

```
(%i18) if r=len then
```

```
    print("Given matrix is diagonalized")
```

```
else
```

```
    print("Given matrix is not diagonalized");
```

Given matrix is diagonalized

```
(%o18) Given matrix is diagonalized
```

```
(%i19) I:matrix([1,0],[0,1]);
```

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

```
(%i20) Char1:charpoly(A,x);
```

```
Char1 (2-x)2 - 1
```

```
(%i21) expand(Char1);
```

```
(%o21) x2 - 4 x + 3
```

```
(%i22) A^^2-4*A+3*I;
```

$$(\%o22) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

```
(%i23) B:matrix([4,0],[0,6]);
```

$$B = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}$$

```
(%i24) eval:eigenvalues(B);
```

```
eval [[4,6],[1,1]]
```

```
(%i25) evvec:eigenvectors(B);
```

```
evvec [[ [4,6],[1,1]], [[ [1,0],[0,1]] ]]
```

```
(%i26) evvecmatrix:matrix([1,0],[0,1]);
```

evcmatrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(%i27) **r:rank(evcmatrix);**

r 2

(%i28) **len:length(evcmatrix);**

len 2

(%i29) **if r=len then**

print("Given matrix is diagonalized")

else

print("Given matrix is not diagonalized");

Given matrix is diagonalized

(%o29) Given matrix is diagonalized

(%i30) **Char2:charpoly(B,x);**

Char2 $(4 - x) (6 - x)$

(%i31) **expand(Char2);**

(%o31) $x^2 - 10x + 24$

(%i32) **B^^2-10*B+24*I;**

(%o32) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(%i40) **C:matrix([1,0,0],[0,2,0],[0,0,3]);**

C $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(%i41) **eval:eigenvalues(C);**

eval $[[1,2,3],[1,1,1]]$

(%i42) **evc:eigenvectors(C);**

evc $\left[\left[[1,2,3], [1,1,1] \right], \left[\left[[1,0,0], [0,1,0], [0,0,1] \right] \right] \right]$

(%i43) **evcmatrix:matrix([1,0,0],[0,1,0],[0,0,1]);**

evcmatrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(%i44) **r:rank(evcmatrix);**

r 3

(%i45) **len:length(evcmatrix);**

len 3

(%i46) **if r=len then**

print("Given matrix is diagonalized")

else

print("Given matrix is not diagonalized");

Given matrix is diagonalized

(%o46) Given matrix is diagonalized

```
(%i47) I:matrix([1,0,0],[0,1,0],[0,0,1]);
```

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
(%i49) char3:charpoly(C,x);
```

```
char3 (1-x) (2-x) (3-x)
```

```
(%i50) expand(char3);
```

```
(%o50)  $-x^3 + 6x^2 - 11x + 6$ 
```

```
(%i51) -C3+6·C2-11·C+6·I;
```

```
(%o51)  $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 
```

```
(%i52) D:matrix([2,1,0],[1,2,0],[0,0,3]);
```

$$D = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

```
(%i53) eval:eigenvalues(D);
```

```
eval [[1,3],[1,2]]
```

```
(%i54) evect:eigenvectors(D);
```

```
evect  $\left[ \left[ \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix} \right], \left[ \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right] \right]$ 
```

```
(%i55) evectmatrix:matrix([1,-1,0],[1,1,0],[0,0,1]);
```

```
evectmatrix  $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 
```

```
(%i56) r:rank(evectmatrix);
```

```
r 3
```

```
(%i57) len:length(evectmatrix);
```

```
len 3
```

```
(%i58) if r=len then
```

```
    print("Given matrix is diagonalized")
```

```
else
```

```
    print("Given matrix is not diagonalized");
```

```
Given matrix is diagonalized
```

```
(%o58) Given matrix is diagonalized
```

```
(%i59) char4:charpoly(D,x);
```

```
char4  $x + (2-x)^2 (3-x) - 3$ 
```

```
(%i60) expand (char4);
```

```
(%o60)  $-x^3 + 7x^2 - 15x + 9$ 
```

```
(%i61) -D3+7·D2-15·D+9·I;
```

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(%i62) **E:matrix**([4,1,0],[0,5,1],[0,0,6]);

$$E = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 6 \end{bmatrix}$$

(%i63) **eval:eigenvalues**(E);

eval [[4,5,6],[1,1,1]]

(%i64) **evect:eigenvectors**(E);

evect $\left[\left[[4,5,6], [1,1,1] \right], \left[[1,0,0], [1,1,0], [1,2,2] \right] \right]$

(%i65) **evectmatrix:matrix**([1,0,0],[1,1,0],[1,2,2]);

$$\text{evectmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 2 \end{bmatrix}$$

(%i66) **r:rank**(evectmatrix);

r 3

(%i67) **len:length**(evectmatrix);

len 3

(%i68) **if** r=len **then**

print("Given matrix is diagonalized")

else

print("Given matrix is not diagonalized");

Given matrix is diagonalized

(%o68) Given matrix is diagonalized

(%i69) **char5:charpoly**(E,x);

char5 $(4-x)(5-x)(6-x)$

(%i70) **expand** (char5);

(%o70) $-x^3 + 15x^2 - 74x + 120$

(%i72) **-E^3+15·E^2-74·E+120·I;**

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$